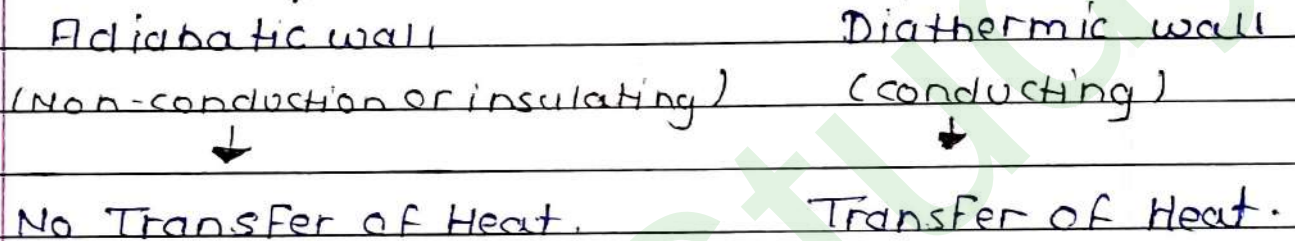


Thermodynamics

- Thermodynamics is the branch of physics that deal with the concepts of heat and temperature and the interconversion of heat and other forms of energy.
- In thermodynamics, we consider that the thermodynamics systems as a whole and learn the interaction of heat and energy during the change of one thermodynamic state to another.
- In this, we study about macroscopic variables like pressure, temperature, volume, internal energy and entropy.
- * Thermal Equilibrium
 - State where all the macroscopic properties of system (P, V, T, M, n) are CONSTANT with time.
 - Two systems when in contact with each other are said to be in thermal equilibrium when their temperatures become the same.

- If two systems are separated by adiabatic process, then despite of the temp. difference both systems will remain in thermal equilibrium separately.

Types of Boundries



>> Zeroth law of thermodynamics

- ⇒ "If two systems are in thermal equilibrium with a third system separately then the two systems are in thermal equilibrium with each other."

- Given by R.H Fowler in 1931 long after 1st, 2nd Law.

State variables / Functions:-

A property whose value does not depend on the path taken to arrive at that specific value is also known as a state function or point function.

Ex. P, V, T , moles (amount of substance) etc.

State Variables can be extensive or intensive.

Path variables / functions:-

Path functions on the other hand, are defined as Function that are dependent on the path between two points.

Only two(s) are! work and Heat.

► Internal Energy

→ Molecular level energy (can be PE, KE, vibrational energy of molecule)

→ Internal energy is the sum of molecular KE and PE in the frame of reference relative to which the centre of mass of system is at rest.

→ U is a state Function

→ U is simply a macroscopic variable.

→ If a system is in rest and have no field, then

[Total energy = T.E of the system]

→ T.E is a State function (like P, V, T)

ΔU only depend upon state (initial and final position) and not on path.

→ In a cyclic process, $\Delta U = 0$, (in state Function)

Note: K.E depends upon the frame of reference

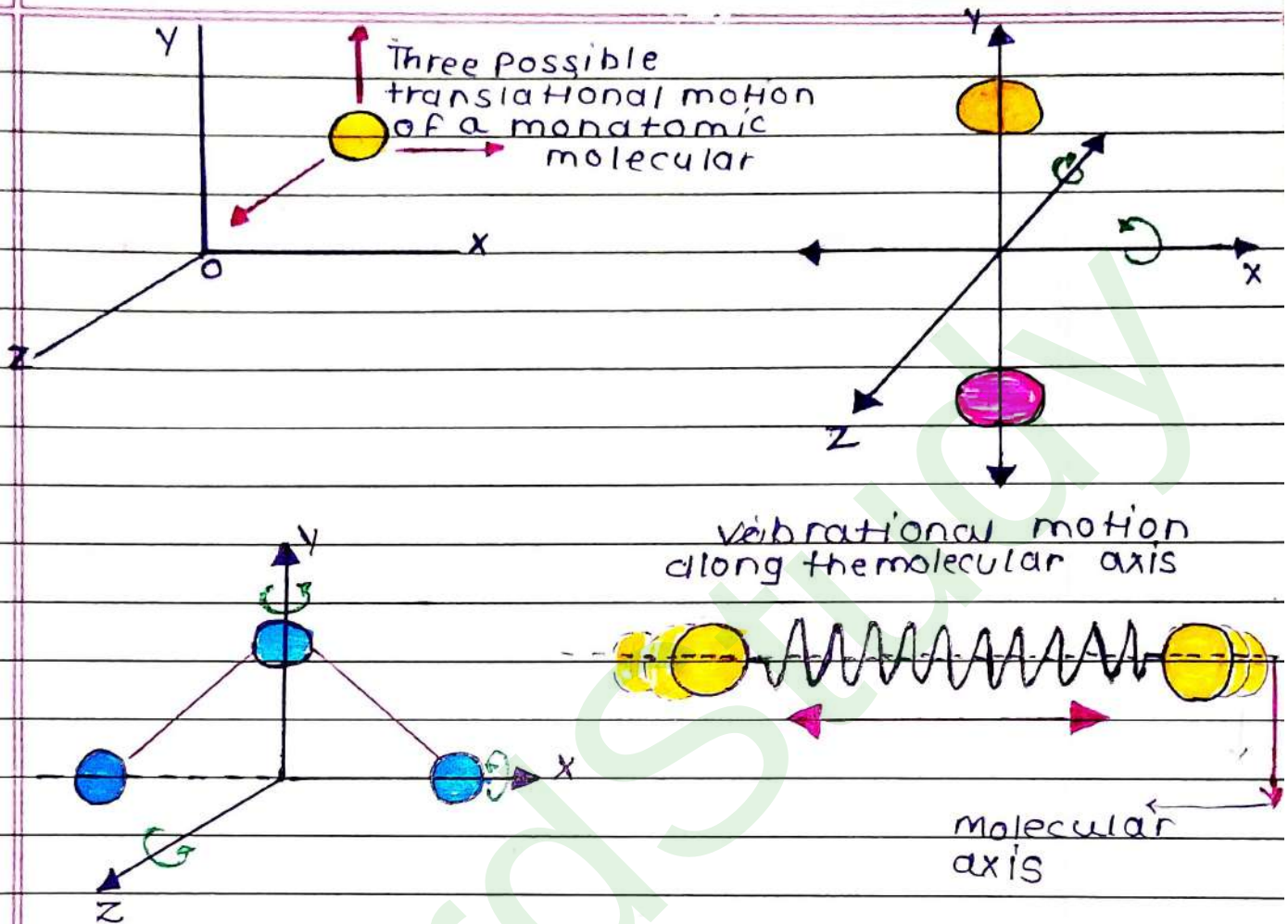
→ ★ For an ideal gas, T.E of Ideal gas = $I.E = K.E$

[U is a function of temperature]

There is no force of attraction among molecules due to which there is no P.E

Degree of freedom

The number of independent ways in which a molecules of gas can move is called the degree of freedom.



- Depends upon Atomicity of Gas

In monoatomic gas, $f = 3$

In diatomic gas, $f = 5$ (3 translational + 2 vibrational)

In polyatomic gas, $f = 6$ ✓ [Linear, $f = 5$]

* At high temp, vibration f is added (+2)

>> Law of Equipartition of Energy

→ "The average KE associated with each molecule with each degree of freedom is $\frac{1}{2} kT$ "

$k \rightarrow$ Boltzman constant ($= 8.314 \text{ J/K}$) ($k = \frac{R}{N}$)

$T \rightarrow$ Absolute temperature

Avogadro's number
↓
Universal Gas constant

★ $U_{\text{molecule}} = n \cdot \frac{f}{2} kT$
(for n molecule)

($R = 8.314 \text{ J/K}$)

★ $U_{\text{ideal gas}} = k \cdot E = n \cdot \frac{f}{2} kT \times N$ ($N \times k = R$)
 n mole

$U_{\text{mole}} = n \cdot \frac{f}{2} RT$

$\Delta U = n \left[\frac{fR}{2} \right] \Delta T$ Always valid $[\Delta U = nC_v \Delta T]$
(Always)

- for ideal gas, U depends only on temperature

★ W and $Q \Rightarrow$ Path function (cyclic $\rightarrow \Delta U = 0$)

ΔU in terms of PV :

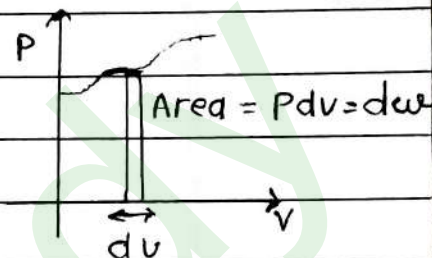
$\left[\Delta U = \frac{f}{2} (P_2 V_2 - P_1 V_1) \right]$

WORK

→ work done $[w] = P\Delta V$ [when P is constant]

→ when P is not constant.

$$\text{Total } w = \int P dV = \text{Total Area}$$



$$w = \text{Area under Graph}$$

$$w = \int P dV \rightarrow \text{Always valid} \Rightarrow \text{work is path dependent}$$

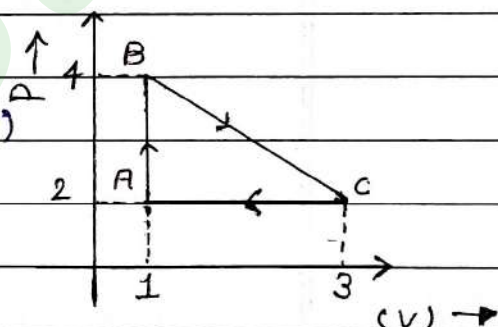
→ $w_{AB} = 0$ (no change in Vol.)

$w_{BC} = \text{Area under BC (Gas Expand)}$

$$= \frac{1}{2} \times (4+2) \times 2 = 6 \text{ atmL}$$

$$w_{CA} = 2 \times 2 = -4 \text{ atmL}$$

(compress)



$$w_{ABCA} = w_{AB} + w_{BC} + w_{CA} \quad (\text{work is not a state function})$$

$$= 0 + 6 - 4$$

$$= 2 \text{ atmL}$$

[work and Heat both are Path dependent]

OR

$$w_{\text{cyclic process}} = \text{Area in cycle} \quad [1 \text{ Joules} = 101.5 \text{ atmL}]$$

$$= \frac{1}{2} \times 2 \times 2$$

$$= 2 \text{ atmL}$$

IF work done is cyclic,

$w = +ve$

(clockwise)

Non-cyclic $w = -ve$

$w = -ve$

(Anticlockwise)

→ main difference in chemistry and physics thermodynamics

	chemistry	Physics
work done by system →	-ve	+ve (Gas expand)
work done on system →	+ve	-ve (contract)

molar Heat capacity (C)

The total amount of energy in the form of heat needed to increase the temperature of 1 mole (and) of any substance by 1 unit is called the molar heat capacity (C) of that substance.

$$\text{Heat exchange (q)} = nC\Delta T$$

molar Heat capacity at constant volume (C_v)

In a system, C_v is the amount of heat energy released or absorbed by a unit mole of the substance with the change in unit temperature at a constant volume.

$$C_v = \left(\frac{f}{2} \right) R$$

where f is Freedom of degree of Gases.

Molar Heat capacity at constant Pressure (C_p)

In a system, C_p is the amount of Heat energy released or absorbed by a unit mole of the substance with the change in unit temperature at a constant pressure.

Mayer's Relation

$$C_p - C_v = R$$

$$\Rightarrow C_p = C_v + R$$

where R is a universal gas constant
Heat capacities Ratio (γ)

$$\gamma = \frac{C_p}{C_v}$$

Summary of f , C_v , C_p and γ

Gas Type	Degrees of Freedom (F)	molar Heat capacity at constant Pressure	molar heat capacity at constant Pressure (C _P)	Heat capacities Ratio $C_V = C_P / C_V$
monoatomic	3 (3 translational)	$2/3 R$	$2/5 R$	$3/5 \approx 1.67$
Diatomic	5 (3 translational, 2 rotational)	$2/5 R$	$2/7 R$	$5/7 = 1.4$
Linear Triatomic	5 (3 translational, 2 rotational)	$2/5 R$	$2/7 R$	$5/7 = 1.4$
Non-linear Triatomic	6 (3 translational, 3 rotational)	$3R$	$4R$	$3/4 \approx 1.33$

HEAT [Q]

- Heat is the exchange of energy between System and surrounding.
- Heat is the form of transfers of energy, when there is temperature different.

- Note:- Heat given to the system $\rightarrow +ve$ } Chemistry
Heat given by the system $\rightarrow -ve$ } Physics

$$Q = nC\Delta T = mS\Delta T$$

Where C = molar heat capacity and
 S = specific heat capacity.

first Law of Thermodynamics

- This law is the sum of many laws which are based on law of conservation of mass.
- It can be applied in any system in which the energy transfer from or to the surroundings.

$$\boxed{\Delta Q = \Delta W + \Delta U}$$

Heat given to the system work done by system Increase in internal energy of system.

Q. given to system $\Rightarrow$ gain = +ve

Q. given by system $\Rightarrow$ loss = -ve

Some Important formulas

$$\odot Q = n C_V \Delta T \quad (V \rightarrow \text{constant}) \quad \odot \Delta V = \frac{n f R \Delta T}{2} = n C_V \Delta T$$

$$\odot Q = n C_P \Delta T \quad (P \rightarrow \text{constant})$$

$$\odot C_V = \frac{f}{2} R \quad (R = 8.314 \text{ J/mol K or } \sim 2 \text{ cal/mole } ^\circ\text{C})$$

$$f = 3, 5, 6$$

monatomic diatomic Polyatomic Gas

$$\odot C_P - C_V = \frac{R}{M} \quad (\text{Mass})$$

$$\odot C_P - C_V = R \quad [\text{Mayer's Equation}] \text{ for any gas (molar)}$$

$$\rightarrow C_P = C_V + R = \frac{f}{2} R + R$$

$$\odot W = P \Delta V \quad (P \rightarrow \text{constant})$$

$$\odot W = \int P dV \quad (P \rightarrow \text{variable})$$

$$\odot C_P = \left(\frac{f}{2} + 1 \right) R$$

$$\odot \gamma = \frac{C_P}{C_V} = 1 + \frac{2}{f} \quad (\text{Ratio of molar heat capacity at constant pressure to constant volume})$$

C_P and C_V in terms of R and γ

$$C_V = \frac{f}{2} R = \frac{R}{\gamma - 1} \Rightarrow \left[C_V = \frac{R}{\gamma - 1} \right], \left[C_P = \frac{\gamma R}{\gamma - 1} \right]$$

Types of Thermodynamics Process

1. Isothermal process

Temperature is constant, $\Delta T = 0$

$PV = \text{constant}$ (Boyle's Law)

$$P_1 V_1 = P_2 V_2$$

In Ideal gas, $[\Delta U = 0]$ for isothermal [slope = $-\frac{P}{V}$]

$$Q = \omega + \Delta U^0$$

$$[Q = \omega]$$

A perfect isothermal process is imaginary.

Conditions:- Process should very-very slow (Quasi-Static)

Boundary should perfectly conductive (diathermic)

→ work done in isothermal process:-

$$\omega = \int_{V_1}^{V_2} P dV$$

At constant, T ideal gas expand

From $V_1, P_1 \rightarrow V_2, P_2$

$$\omega = \int_{V_1}^{V_2} nRT dV = nRT \int_{V_1}^{V_2} \frac{dV}{V}$$

work done
by the gas

$$= nRT [\log_e V]_{V_1}^{V_2} = nRT [\log V_2 - \log V_1]$$

$$= nRT \log \left[\frac{V_2}{V_1} \right]$$

$$W = nRT \log \left[\frac{V_2}{V_1} \right] = nRT \log \left[\frac{P_1}{P_2} \right] = P_1 V_1 \text{ or } P_2 V_2 \log \left[\frac{P_1}{P_2} \right]$$

$$P_1 V_1 = P_2 V_2$$

$$P_1 V_1 = nRT$$

$$\left(\frac{P_1}{P_2} = \frac{V_2}{V_1} \right)$$

$$P_2 V_2 = nRT$$

- molar S.H.C = ∞
- $\Delta U = 0$ (For ideal gas)

2. Adiabatic Process

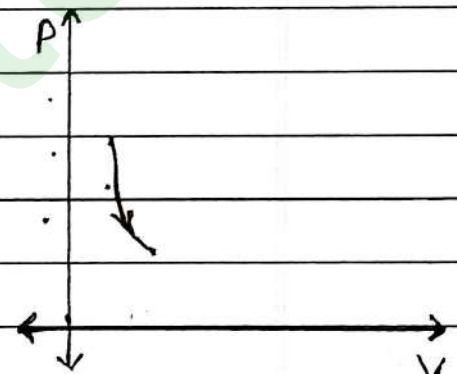
- No Heat exchange, $Q = \Delta Q = 0$
 $Q = W + \Delta U$

$$[W = -\Delta U]$$

work done by system

- Also a imaginary process
 condition - (i) Process should
 very-very fast

(ii) Perfectly insulating
 wall



$$\left[\text{slope} = -\gamma \left(\frac{P}{V} \right) \right]$$

$$\left(\frac{dP}{dV} \right)$$

slope depends on
 γ

$$\gamma_{\text{mono}} > \gamma_{\text{dia}} > \gamma_{\text{tria}}$$

$$\text{Relation, } [PV^\gamma = \text{constant}]$$

$$\text{slope}_{\text{mono}} > \text{slope}_{\text{dia}} > \text{slope}_{\text{tria}}$$

(decrease faster)

$$P_1 V_1^\gamma = P_2 V_2^\gamma \rightarrow \text{poisson's equation}$$

in term of T & V

$$PV^\gamma = \text{constant}$$

$$\frac{nRT}{V} V^\gamma = \text{constant}$$

$$TV^{\gamma-1} = \text{constant}$$

$$P \left(\frac{nRT}{P} \right)^{\gamma} = \text{constant}$$

$$P^{1-\gamma} T^{\gamma} = \text{constant}$$

work done in Adiabatic Process

$$W = -\Delta U$$

$$C_p - C_v = R, \frac{C_p}{C_v} = \gamma$$

$$\Delta U = n C_v \Delta T$$

$$\Delta U = \frac{nR}{\gamma-1} (T_2 - T_1)$$

$$\gamma(C_v - C_v) = R$$

$$C_v (\gamma - 1) = R$$

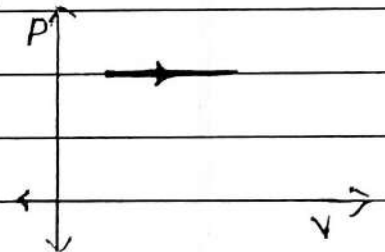
$$W = \frac{nR}{\gamma-1} (T_2 - T_1)$$

$$(C_v = \frac{R}{\gamma-1})$$

$$\Delta Q_p = n C_p \Delta T, \Delta U = n C_v \Delta T$$

3. Isobaric $[W = \Delta Q_p - \Delta U = nR\Delta T]$

Pressure is constant



$$W = \int_{V_1}^{V_2} P dV = P [V]_{V_1}^{V_2} = P(V_2 - V_1) = P\Delta V$$

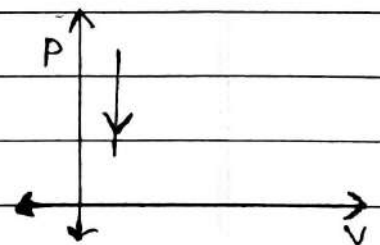
$$[W = P\Delta V]$$

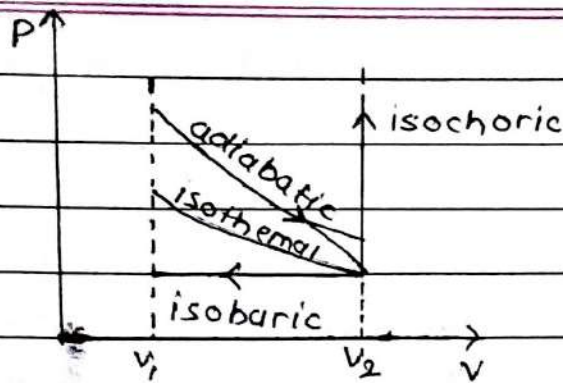
4. Isochoric

$V \rightarrow \text{constant}$

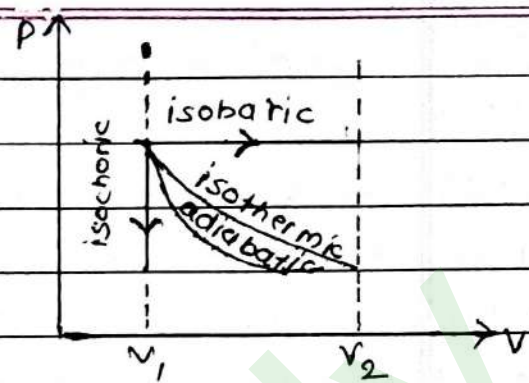
$$[W = 0]$$

$$[Q = \Delta U]$$





compression
 $w_{adi} = \max$



expansion
 $w_{isobaric} \rightarrow \max$

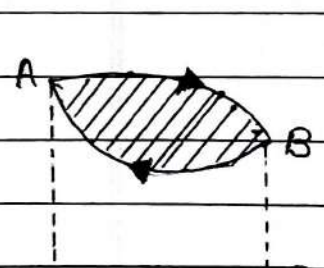
Cyclic Process

A cyclic Process refers to the Process wherein the system returns to its initial stage after undergoing a series of changes.

Example Indicator Diagram

Here, change in internal energy, ΔU

work done, $w = \text{Area under the loop}$,



$\Rightarrow Q = W$ according to the First law of thermodynamics.

Also, W is positive when the cycle is Clockwise whereas it is negative when the cycle is anti clockwise.

Second law of Thermodynamics

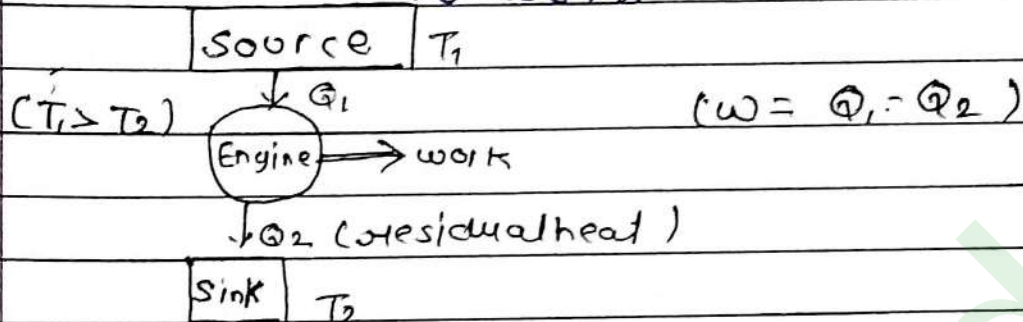
There are a lot of ways in which this law can be stated. Even though all the statements are the same with respect to their contents, the following two are the most significant.

Kelvin Planck statement:- No process is possible whose sole result is the absorption of heat from a reservoir and the complete conversion of the heat into work.

Clausius Statement:- No process is possible whose sole result is the transfer of heat from a colder object to a hotter object.

- 1st Law does not tell whether process will happen or not.
- 2nd Law is a natural law which talks about spontaneous and non-spontaneous process.
- 'All work can be converted to heat but all heat cannot be converted to work.
- Efficiency of a heat engine and the co-efficient of performance of a refrigerator are explained by 2nd Law.
- Reversible process - could happen again
Irreversible process - could not happen/ occur again.

[Heat Engine] - Device which converts heat to work



Efficiency of heat engine = $\frac{\text{work done}}{\text{Heat given}} \times 100$

$$\left[\eta = \frac{W}{Q_1} \times 100 = \frac{Q_1 - Q_2}{Q_1} \times 100 \right]$$

$$\left[\eta = \left(1 - \frac{Q_2}{Q_1} \right) \times 100 \right] \quad \text{For any engine.}$$

→ η can never be 1 or 100%.

→ Heat engine gives cyclic graph for any type.

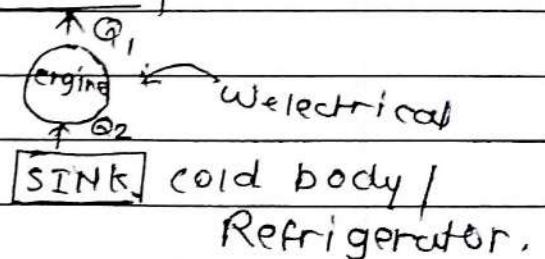
Refrigerator

- with the help of electrical energy (work) → extract heat from cold body

Source Room.

Reverse of Heat engine

$$\left(\frac{Q_2}{Q_1} = \frac{T_2}{T_1} \right)$$



Coefficient of performance (κ) or (α)

$$\left[\alpha = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2} \right]$$

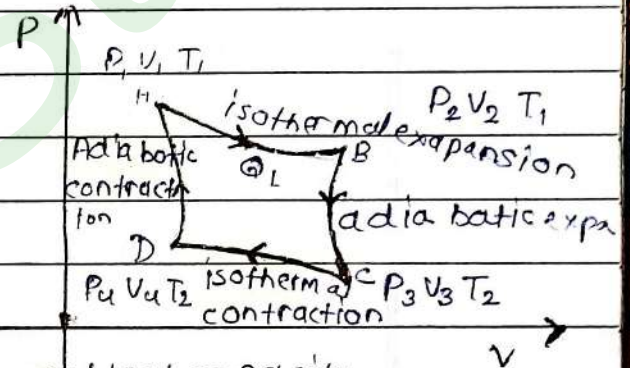
α = Heat Reject $\uparrow$, W $\downarrow$

► Carnot's Heat Engine

- Imaginary engine
- η = max. of any such engine which works b/w same temp. T_1 & T_2

$$\left[\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} \right]$$

All process in this are reversible



∞ Small weights

∞ Heat capacity

∞ Heat capacity

Perfectly insulating

perfectly conducting

(Perfectly conducting)

(Perfectly conducting)

perfectly insulating

Frictionless Piston

T_1

T_2

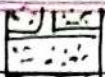
Stand

Source

Sink

Stand

Step 1



• slowly remove weight (isothermal expansion)

Q₁

Source

• Temp. constant, gas expand

Step 2



• weight remove very fast (adiabatic expansion)

Stand

• until temp. reaches to T₂

Step 3



• Put weight again slowly (isothermal contraction)

Q₂

Sink

• -w (heat release)

Step 4



• Put weight fastly (adiabatic compression)

Stand

⇒ If there is a heat engine which is perfectly reversible and which has no loss of heat other than rejection working b/w temperature.

T₁ and T₂ it will always $\eta = 1 - \frac{T_2}{T_1}$ and no engine will have η more than this in temperature $\cdot T_1$ and T₂

$\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$	(in K)